Massachusetts Institute of Technology
Department of Electrical Engineering and Computer Science
6.3800 Introduction to Inference
Fall 2026

Recitation 3 Handout

Thursday, September 17

Problem 3.1

Of the students in a class, 10 are freshmen, 30 are sophomores, 20 are juniors and 10 are seniors. Let X be the year of a student chosen at random in the class.

(a) Determine the alphabet $\mathcal{X}$ and the PMF p_X for the random variable X.

(b) Determine the mean, variance, and entropy of X.

(c) Suppose each freshman takes 4 courses each term, and all other students take 5 courses each term. Determine the expected number of courses a student chosen at random from the class is taking this semester.

(d) If 10 more seniors join the class, will the entropy of X increase of decrease?

Problem 3.2

Bob has a biased coin that turns up heads with probability p. He tosses the coin repeatedly until he observes heads. Assume that coin tosses are independent of each other. Let X be the number of coin tosses up to and including the toss that produced heads.

(a) Determine and sketch $p_X(x)$ for $x = 1, 2, \ldots$.

(b) Show that $\sum_{x=1}^{\infty} p_X(x) = 1$.

Hint: Recall that for $|r| < 1$,

$$\sum_{i=0}^{\infty} r^i = \frac{1}{1-r},$$

which is referred to as a geometric series.

(c) Show that for any random variable $Y \in \{0, 1, 2, \ldots\}$, $\mathbb{E}[Y] = \sum_{y=1}^{\infty} \mathbb{P}(Y \geq y)$.

(d) Show that $\mathbb{E}[X] = 1/p$.

Hint: You may find the result in part (c) helpful.